

VELMÈRE PRO AUDIT REPORT

SafeMoon Protocol Core (0x8076c74c5e3f5852037f31ff0093eeb8c8add8d3)

VELMERE SECURITY AUDIT REPORT: SAFEMOON PROTOCOL CORE

Network: BNB Smart Chain (BSC) (Chain ID: 56)

Report ID: rep_safemoon_pro_047 | Tier: PRO | Application Surface: CANONICAL

Created At: 2026-09-10T12:36:16.120Z

VERDICT SUMMARY:

CRITICAL RISK (Risk: 88/100 | Quality: 83/100)

Release Decision: BLOCKED | Verification Status: PARTIALLY_VERIFIED

STOP-SELL DIRECTIVE: CRITICAL_RISK_THRESHOLD: Score exceeds maximum institutional tolerance

Confidence Score: 99/100

Evidence Coverage: 97%

EVM State Provenance: Block #31500000 | Bytecode SHA-256: sha256:88771122a... | DETERMINISTIC_REPRODUCIBLE

VELMERE CRYPTOGRAPHIC AUDIT SEAL - SHA-256 MERKLE ROOT

[VERIFIED - IMMUTABLE]

Root: sha256:6ca6f93be1e0f46b...20891f073e40



Standard: Merkle Non-Repudiation Seal | RFC 3161 Pinned | Verification: /api/audit/report-pdf

Evidence Coverage Matrix: Bytecode 91% | CFG 86% | Functions 94% | Detectors 100% | Formal Verification 76%

CRITICAL RISK: Legacy SafeMoon v1 contract (0x8076...8add8d3, migrated to v2 0x4298...fcb5). Exhibits centralized owner privilege vectors including arbitrary fee adjustments up to 100% without hard caps. Historical incident: On March 28, 2023, public/unrestricted burn functionality was exploited to arbitrarily burn tokens held by the liquidity pair, artificially inflating the price and draining ~\$8.9M WBNB in the same transaction.

AUDIT OVERVIEW & CONTRACT CONTEXT

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: SafeMoon Protocol Core

Contract Address: 0x8076c74c5e3f5852037f31ff0093eeb8c8add8d3

Network / Chain ID: BNB Smart Chain (BSC) (ID: 56)

Token / Symbol: SAFEMOON

Proxy Pattern: Custom Proxy with Unchecked Owner Migration

Compiler Version: solc 0.6.12

SOURCE CODE VERIFICATION & BYTECODE DECOMPIlation

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

BASELINE SECURITY FINDINGS

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

Findings & Vulnerability Analysis:

CRITICAL

VLM-SAFE-01 [SWC-105 / CWE-284]: Public / Unrestricted Burn LP Reserve Desynchronization (\$8.9M Exploit)

Category: Liquidity Pool Manipulation / Arbitrary Burn Vector | State: recommended

Evidence: burn() allows targeting arbitrary holders including Uniswap/PancakeSwap pair contracts

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Recommendation: Quarantine legacy v1 contract; enforce pair address exemption from token destruction and use migrated v2 architecture with immutable liquidity timelocks.

HIGH

VLM-SAFE-02 [SWC-105 / CWE-284]: Centralized Privileged Parameter Risk (Uncapped Fee Setter)

Category: Centralized Governance & Fee Parameter Risk | State: recommended

Evidence: function setTaxFeePercent(uint256 taxFee) external onlyOwner { _taxFee = taxFee; }

Recommendation: Revoke or lock fee setter to maximum 3% hard ceiling.

PRIVILEGED ROLES, ADMIN KEYS & GOVERNANCE

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Verification of timelock delay parameters and governance controls

* Owner Multi-sig	Single Private Key (No Multi-sig)	FLAGGED
* Timelock Enforcement	None (Zero Hours Delay)	FLAGGED
* Honeypot Vector	Detected (Exclusion from rewards & fees)	FLAGGED

Detailed role map traces all execution paths back to verifiable on-chain controllers.

Findings & Vulnerability Analysis:

CRITICAL

VLM-SAFE-P01: EOA Controlled Privileged Functions

Category: Access Control | State: open

Evidence: address public _owner; (points to 0xcd1972...)

Recommendation: Blacklist address across automated routing systems.

LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Token distribution curve, whale clusters, and verifiable DEX lock proofs

* Liquidity Vulnerability	LP tokens unlocked or under owner EOA control	FLAGGED
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Liquidity analysis examines AMM pairing architecture, lock mechanisms, and transaction fee behavior.

ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Simulation of oracle manipulation, flash loans, and cross-function reentrancy

* Oracle Dependency	Pricing feeds inspected in bytecode	VERIFIED
* Flash Loan Attack Vector	State manipulation resilience assessed	VERIFIED
* Reentrancy Protection	CEI pattern / Reentrancy guard verified	VERIFIED

STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED)

ADVANCED

[LOCKED] [LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)

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[LOCKED]

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Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): 76% invariants verified (QF_LIA UNSAT)
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)

ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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